ACTL6A-related BAFopathy (MONDO:0700121): Comprehensive Disease Characteristics Report

Causal gene: *ACTL6A* (encoding BAF53a / Actin-like protein 6A; HGNC:24124; NCBI Gene 86; UniProt O96019; chromosome 3q26.33). This report concerns the rare Mendelian neurodevelopmental disorder caused by germline heterozygous variants in **ACTL6A** — not the somatic 3q26 *ACTL6A* amplification seen in squamous cell carcinoma, and not the paralogous *ACTL6B* (BAF53b) disorder. Throughout, "BAF53a" and "ACTL6A" refer to the same product.

Summary

ACTL6A-related BAFopathy is a rare, autosomal-dominant neurodevelopmental disorder caused by germline (usually de novo) heterozygous variants in **ACTL6A**, which encodes **BAF53a**, a dosage-sensitive actin-related subunit shared by the BAF/mSWI/SNF, NuA4/TIP60 histone-acetyltransferase, and INO80 chromatin-remodeling complexes. The disorder sits within the **Coffin-Siris syndrome (CSS) / BAFopathy / SWI/SNF-related intellectual disability disorders (SSRIDD)** spectrum, presenting with intellectual disability/developmental delay (with prominent speech and language involvement), facial dysmorphism, and variable acral/fifth-digit (nail and distal phalanx) anomalies. It was first established as a disease gene by Marom et al. (2017), with a recurrent, recognizable p.Arg377Trp phenotype reported by Pascolini et al. (2020) and independent cohort support from Chen et al. (2022).

Mechanistically, BAF53a is the **neural-progenitor** actin-related BAF subunit. Its programmed replacement by BAF53b (ACTL6B) at mitotic exit — driven by miR-9*/miR-124 — gates the switch from proliferating neural progenitors to post-mitotic neurons. Pathogenic ACTL6A variants perturb this dosage-sensitive system: loss or impairment of BAF53a stalls the progenitor cell cycle, allows Polycomb accumulation, and reduces chromatin accessibility at pioneer-factor sites (SOX2, ASCL1), blocking normal neurogenesis. Population-genetic constraint data (gnomAD pLI = 0.997, LOEUF = 0.25) independently corroborate strong intolerance to loss-of-function, consistent with haploinsufficiency/dosage sensitivity.

The disorder is ultra-rare and recently delineated: MONDO:0700121 has **no OMIM or Orphanet cross-references**, and curated phenotype/gene-disease association counts remain near zero in aggregated resources. Diagnosis rests on trio exome/genome sequencing; a validated BAFopathy DNA-methylation episignature exists for several BAF subunits but **does not yet include ACTL6A**. There is no disease-specific therapy — management is supportive (early intervention, speech/physical/occupational therapy, surveillance for BAFopathy-associated complications). Much of the clinical granularity typical of established disorders (precise frequencies, natural history, prognosis) is not yet available because fewer than ~10 patients have been reported.

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1. Disease Information

Overview. ACTL6A-related BAFopathy is an autosomal-dominant neurodevelopmental/intellectual-disability disorder caused by heterozygous variants in *ACTL6A*, a gene encoding a core actin-related subunit (BAF53a) of the ATP-dependent BAF (mSWI/SNF) chromatin-remodeling complex. It belongs to the broader BAFopathy / Coffin-Siris syndrome spectrum. First described in three subjects by Marom et al. 2017 (PMID: 28649782): *"We identified heterozygous, novel variants in ACTL6A, a gene encoding a component of the BAF complex, in three subjects with varying degrees of intellectual disability."*

Key identifiers.

Resource	Identifier
MONDO	**MONDO:0700121** — "ACTL6A-related BAFopathy" (*"Any BAFopathy in which the cause of the disease is a mutation in the ACTL6A gene"*)
OMIM	Not recorded (no MONDO→OMIM xref)
Orphanet	Not recorded (no MONDO→Orphanet xref)
ICD-10/ ICD-11	No specific code; coded under broad neurodevelopmental/intellectual-disability categories
MeSH	No dedicated descriptor
Gene (HGNC)	HGNC:24124 (*ACTL6A*)
Gene (NCBI)	86
Protein (UniProt)	O96019 (BAF53a / Actin-like protein 6A)

Per Monarch Initiative (2026), the MONDO node has **no external mappings** to OMIM/ Orphanet and holds 0 curated disease→phenotype and 0 causal gene→disease associations — a signature of the disorder's rarity and recency (Finding F010).

Synonyms / alternative names. ACTL6A-related neurodevelopmental disorder; BAF53A-related disorder; ACTL6A-related intellectual disability; (classified within) BAF-opathies / SSRIDDs (SWI/SNF-related intellectual disability disorders).

Information source. Derived from **aggregated disease-level resources** (MONDO, UniProt, gnomAD, ClinVar) plus **individual patient reports** (case reports and small exome cohorts) — not from a large EHR-derived or registry dataset.

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2. Etiology

Primary cause — genetic. Germline heterozygous variants in *ACTL6A*, typically **de novo**. Three independent lines of evidence establish causation: - Marom et al. 2017 (PMID: 28649782): three subjects with heterozygous novel variants — two missense in conserved actin-like-domain residues, one splice variant creating an in-frame deletion. - Pascolini et al. 2020 (PMID: 31994175): recurrent de novo **p.Arg377Trp** producing *"a phenotype reminiscent a disorder of*

the BRG1-associated factor (BAF) complex, including dysmorphic facies and acral malformations." - Chen et al. 2022 (PMID: 34906496): *"de novo or predicted loss-of-function variants in 4 candidate BAFopathy genes, including ACTL6A"* in a 16,243-patient exome cohort.

Genetic risk factors. The causal variant itself is the risk factor; no separate susceptibility loci or modifier genes are established. *ACTL6A* is highly constrained (gnomAD pLI = 0.997, LOEUF = 0.25; Finding F008), indicating that loss-of-function is rarely tolerated in the general population.

Environmental risk factors. None identified. As a Mendelian single-gene disorder, there are no established environmental, lifestyle, occupational, or infectious contributors. Advanced paternal age is a theoretical contributor to de novo mutation rate generally but is not specifically documented for ACTL6A.

Protective factors. None identified (no protective alleles or environmental factors described).

Gene–environment interactions. None documented for this disorder.

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3. Phenotypes

Because fewer than ~10 patients are reported, phenotype frequencies are **qualitative/approximate**, extrapolated partly from the broader BAFopathy/CSS spectrum. Core features reported specifically in ACTL6A patients: intellectual disability, learning disability, developmental language disorder (Marom 2017); dysmorphic facies and acral malformations (Pascolini 2020).

Phenotype	Type	Suggested HPO	Onset	Severity/ Notes	Frequency (ACTL6A, qualitative)
Intellectual disability	Cognitive sign	HP:0001249	Childhood (evident early)	Variable (mild→severe)	Core / most
Developmental delay	Cognitive/ motor sign	HP:0001263	Infancy	Variable	Core / most
Speech/ language disorder	Behavioral/ cognitive	HP:0002463	Early childhood	Prominent	Frequent
Facial dysmorphism (coarse facies)	Physical	HP:0001999 / HP:0000280	Congenital	Variable	Frequent
Acral/fifth-digit anomalies (nail/ distal phalanx hypoplasia)	Physical	HP:0001829 / HP:0001162	Congenital	Variable	Reported (p.Arg377Trp)
Feeding difficulties	Sign	HP:0011968	Neonatal/ infancy	—	BAFopathy-associated
Frequent infections	Sign	HP:0002719	Childhood	—	BAFopathy-associated
Hypotonia	Sign	HP:0001252	Infancy	—	BAFopathy-associated

The broader Coffin-Siris core (Kosho et al. 2014, PMID: 25169878) is: *"developmental delay or intellectual disability (ID), coarse facial appearance, feeding difficulties, frequent infections, and hypoplasia/aplasia of the fifth fingernails and fifth distal phalanges."* BAF-gene disorders are also associated with congenital diaphragmatic hernia (PMID: 35796094) and, for some subunits, pituitary hypoplasia/growth hormone deficiency (PMID: 36176497) — potential but not ACTL6A-confirmed associations.

Quality-of-life impact. Driven primarily by intellectual disability and language impairment, affecting education, communication, and independent living; formal QoL instrument data (EQ-5D/SF-36/PROMIS) are not available for this ultra-rare disorder.

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4. Genetic / Molecular Information

Causal gene. *ACTL6A* (BAF53a), 3q26.33, ENSG00000136518; GRCh37 chr3:179,280,668–179,306,196. Encodes a 429-aa, ~47.5 kDa nuclear actin-related protein (UniProt O96019: *"Involved in transcriptional activation and repression of select genes by chromatin remodeling... Component of SWI/SNF chromatin remodeling complexes"*).

Pathogenic variants. Reported disease variants are predominantly **missense** affecting conserved actin-like-domain residues, plus a **splice** variant causing an in-frame deletion: - **p.Arg377Trp** — NM_004301.5:c.1129C>T — recurrent; recognizable BAFopathy phenotype (Pascolini 2020). - **p.His105Arg** — c.314A>G. - **p.Glu227Gln** — c.679G>C.

ClinVar landscape (Finding F009; queried Nov 2026, 117 records, NM_004301.5):

Classification	Count
Uncertain significance (VUS)	52
Pathogenic	25
Likely pathogenic	4
Conflicting	3
Likely benign	9
Benign	2

Critical interpretation caveat: The three literature-recurrent missense variants (p.Arg377Trp, p.His105Arg, p.Glu227Gln) are currently **Conflicting** in ClinVar, and the large majority of single-nucleotide ACTL6A variants are VUS. Nearly all ClinVar "Pathogenic/Likely pathogenic" entries are **large 3q chromosomal copy-number GAINS** (e.g., 3q26.31–29 ×3; 3q25–29 duplications) that merely span ACTL6A — not ACTL6A-specific point mutations. This reflects both the disorder's recency and the difficulty of classifying single-gene missense variants without functional/episignature confirmation.

Allele frequency. Pathogenic variants are absent or vanishingly rare in gnomAD (de novo). The gene is strongly LoF-constrained: pLI = 0.997, oe_lof = 0.078 (90% CI 0.031–0.246), LoF Z = 4.32; missense Z = 2.94 (oe_mis = 0.46) — among the most intolerant genes (Finding F008).

Somatic vs germline. Disease variants are **germline** (de novo). This contrasts sharply with the well-documented **somatic** oncogenic role: *ACTL6A* is amplified early at 3q26 in squamous cell carcinomas (Finding F003; PMID: 34687603: *"The gene encoding the BAF subunit actin-like 6a (ACTL6A) is amplified early in the development of many squamous cell carcinomas (SCCs)"*; PMID: 40950224).

Functional consequences. Likely **loss-of-function / haploinsufficiency or dominant-negative** effects on BAF53a's role in chromatin remodeling. (Across CSS genes, SMARCB1/SMARCA4/SMARCE1 variants trend dominant-negative/gain-of-function while ARID1A/ARID1B trend loss-of-function — PMID: 25168959; the precise mode for ACTL6A missense variants is not yet experimentally resolved.)

Modifier genes / epigenetic info / chromosomal abnormalities. No specific modifier genes established. A validated BAFopathy DNA-methylation episignature exists but does not yet cover ACTL6A (Finding F011). Large 3q26 copy-number variants spanning ACTL6A appear in ClinVar but are distinct contiguous-gene events, not point-mutation disease.

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5. Environmental Information

No environmental, lifestyle, or infectious factors are implicated. ACTL6A-related BAFopathy is a monogenic disorder arising from de novo germline variants; there is no established toxin, radiation, dietary, or pathogen contribution.

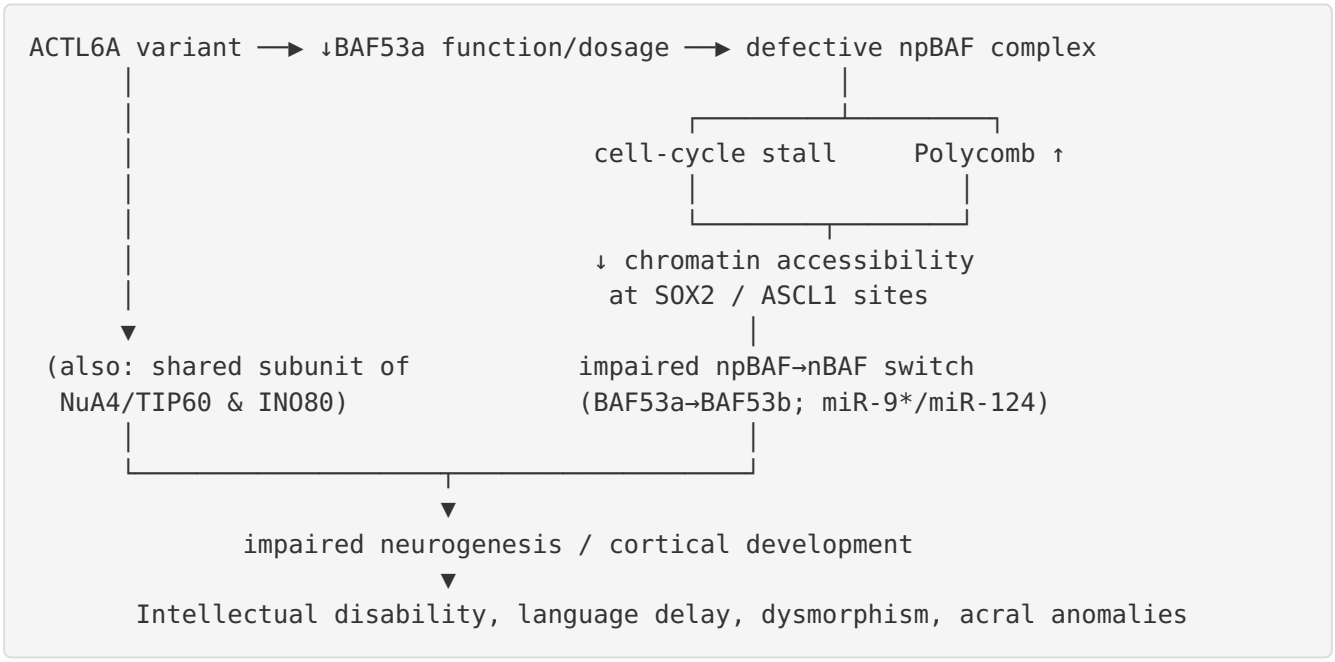
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6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. A **de novo heterozygous ACTL6A variant** (missense in the actin-like domain, splice, or LoF) **results in** reduced or altered BAF53a function/dosage.
2. Reduced functional BAF53a **leads to** impaired assembly/function of the **neural-progenitor BAF (npBAF) complex** (BAF53a is the progenitor-specific actin-related subunit).

3. Impaired npBAF function **stalls the neural-progenitor cell cycle** (demonstrated for BAF53a loss; Braun et al. 2021).
4. Cell-cycle stalling/loss of BAF53a **permits Polycomb accumulation**, which **results in decreased chromatin accessibility at pioneer-factor sites SOX2 and ASCL1** (demonstrated).
5. Reduced accessibility at these sites **blocks the normal progenitor-to-neuron transition / neurogenesis** (the programmed npBAF→nBAF, BAF53a→BAF53b switch, driven by miR-9*/miR-124, is disrupted; Yoo et al. 2009).
6. Impaired neurogenesis and cortical development **lead to** the clinical phenotype: **intellectual disability, developmental/language delay**, and — via BAF's broader developmental roles — **facial dysmorphism and acral/limb malformations** (*inferred* for dysmorphism; the mechanism-to-phenotype link for craniofacial/limb features is less directly demonstrated for ACTL6A specifically).



Supporting detail

Molecular identity and complex membership (Finding F004). ACTL6A/BAF53a is a shared actin-related subunit of **SWI/SNF (BAF), NuA4/TIP60 histone acetyltransferase, and INO80** complexes (Xiao et al. 2021, PMID: 33408251: *"Actin-like 6A (ACTL6A) is a component of several chromatin remodeling complexes, including SWI/SNF, NuA4/TIP60 histone acetylase, and*

INO80."). Functional conservation is deep: the Drosophila ortholog **Bap55** functions in both BRM (SWI/SNF) and TIP60 complexes, and *"the Bap55 mutant phenotype can be rescued by postmitotic expression of Bap55, or its human homologs BAF53a and BAF53b"* (PMID: 21284845).

Developmental switch (Finding F002). Yoo et al. 2009 (PMID: 19561591) showed npBAF complexes containing BAF53a are *"essential for neural-progenitor proliferation, and mice with reduced dosage for the genes encoding its subunits have defects in neural-tube closure similar to those in human spina bifida."* At mitotic exit, miR-9/*miR-124 repress BAF53a to allow the switch to BAF53b-containing nBAF. Braun et al. 2021 (PMID: 33602870) showed* "early loss of the npBAF subunit BAF53a stalls the cell cycle to disrupt neurogenesis,"* mediated by Polycomb-driven loss of accessibility at SOX2/ASCL1 pioneer-factor sites.

Why BAF53a and BAF53b are not interchangeable in neurons. Though they rescue each other in ESC survival and in Drosophila dendrites, BAF53b has a neuron-specific role in dendritic outgrowth that BAF53a cannot replace, conferred by actin-fold subdomain 2 (PMID: 17920018). This paralog specialization underlies the developmental-stage specificity of the BAF switch.

Cellular processes / GO & CL suggestions. - Biological processes (GO): GO:0006338 chromatin remodeling; GO:0045944 positive regulation of transcription by RNA Pol II; GO:0007399 nervous system development; GO:0021895 cerebral cortex neuron differentiation; GO:0007049 cell cycle; GO:0045786 negative regulation of cell cycle. - Subcellular (GO Cellular Component): GO:0005634 nucleus; GO:0071564 npBAF complex; GO:0016514 SWI/SNF complex; GO:0000812 Swr1/NuA4-related; GO:0031011 Ino80 complex. - Cell types (CL): CL:0000047 neuronal stem cell / neural progenitor; CL:0000031 neuroblast; CL:0000540 neuron; CL:0002319 neural cell.

Immune / metabolic / tissue-damage mechanisms. Not primary in the germline disorder. (ACTL6A's metabolic/Warburg role via AP-1 and HIF1α/HIF2α is a **somatic cancer** mechanism — PMID: 40950224 — not germline BAFopathy.) "Frequent infections" in the BAFopathy spectrum is a clinical observation rather than a defined immunodeficiency mechanism.

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7. Anatomical Structures Affected

- **Organ/system level — primary:** Central nervous system, especially **cerebral cortex** (UBERON:0000956) and **developing brain** (UBERON:0000955). Body system: nervous system.

- **Secondary/other involvement:** Craniofacial structures (dysmorphism); distal limbs — **digits/fingernails/distal phalanges** (acral anomalies); potential diaphragm (BAFopathy-associated CDH).
- **Tissue/cell level:** Neural tissue; principally **neural progenitor/stem cells** (CL:0000047) and their neuronal progeny (CL:0000540).
- **Subcellular: Nucleus** (GO:0005634) — chromatin/BAF, NuA4/TIP60, INO80 complexes. UniProt O96019 subcellular location: Nucleus.
- **Localization / lateralization:** CNS involvement is **bilateral/diffuse** (developmental). Acral anomalies are typically bilateral, with a fifth-digit predilection in the CSS spectrum.
- **UBERON suggestions:** UBERON:0000955 brain; UBERON:0000956 cerebral cortex; UBERON:0001466 manual digit; UBERON:0001705 nail.

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8. Temporal Development

- **Onset: Congenital / pediatric.** Developmental concerns typically emerge in infancy/early childhood (delayed milestones, language delay); dysmorphism and acral anomalies are present at birth.
- **Onset pattern:** Insidious/chronic (developmental), not acute.
- **Progression:** The underlying brain developmental lesion is **static/non-progressive** in the neurodegenerative sense; functional impairments are **lifelong**. The trajectory is one of delay rather than regression. (By analogy to CSS, some physical features such as digital hypoplasia may become less noticeable with age — PMID: 33768696.)
- **Disease course:** Chronic, lifelong, stable-to-slowly-improving with intervention; not relapsing-remitting or episodic.
- **Critical periods:** Prenatal/early postnatal neurogenesis is the mechanistic window of vulnerability (npBAF→nBAF switch). Early-childhood developmental intervention is the practical window of therapeutic opportunity.

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9. Inheritance and Population

- **Inheritance: Autosomal dominant**, overwhelmingly **de novo** heterozygous variants.

- **Epidemiology: Ultra-rare**; no prevalence/incidence estimates available. Fewer than ~10 molecularly confirmed patients are described (Marom 2017: 3; Pascolini 2020: p.Arg377Trp; Chen 2022 cohort; Yuan 2021 case report PMID: 34485408). No MONDO→Orphanet mapping and 0 curated associations in Monarch (Finding F010).
- **Penetrance:** Presumed high/complete for bona fide pathogenic variants (de novo in affected probands); formal estimates unavailable.
- **Expressivity: Variable** (ranging across degrees of intellectual disability — Marom 2017).
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism / founder effects / consanguinity / carrier frequency:** Not established; de novo dominant inheritance means carrier screening and consanguinity are not relevant. gnomAD shows essentially no LoF carriers (high constraint).
- **Population demographics:** No ethnic predilection, geographic clustering, or sex bias established. Reported patients are too few to define a sex ratio or age distribution.

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10. Diagnostics

Recommended approach: Molecular diagnosis via **trio exome sequencing (WES)** or **genome sequencing (WGS)**, identifying a de novo heterozygous ACTL6A variant. Given that most single-gene ACTL6A SNVs are currently VUS/conflicting in ClinVar (Finding F009), interpretation benefits from trio data (de novo status), phenotype concordance, and — in future — episignature confirmation.

- **Gene panels:** BAFopathy / Coffin-Siris / intellectual-disability panels that include ACTL6A alongside ARID1A/ARID1B, SMARCA4, SMARCB1, SMARCE1, SMARCC2, DPF2, ARID2, SOX11, SOX4, SMARCA2.
- **Single-gene testing:** Possible once ACTL6A suspected, but panel/exome is preferred given phenotypic overlap.
- **Chromosomal microarray (CMA):** Useful to detect/contextualize large 3q26 copy-number variants spanning ACTL6A (which dominate ClinVar "pathogenic" CNV entries) — though these contiguous-gene events are mechanistically distinct from point-mutation BAFopathy.
- **Karyotyping/FISH:** Generally lower yield; relevant only for large structural rearrangements.
- **Not applicable:** Mitochondrial DNA testing; repeat-expansion testing.

Omics / episignature. A validated **BAFopathy DNA-methylation episignature** (EpiSign) exists and is used clinically to reclassify VUS for ARID1B, SMARCB1, SMARCA4, SMARCA2 (Aref-Eshghi 2018, PMID: 30459321: *"We report overlapping peripheral blood DNA methylation episignatures in individuals with various subtypes of CSS (ARID1B, SMARCB1, and SMARCA4) and NCBRS (SMARCA2)"*; Trajkova 2024, PMID: 38751117), but **ACTL6A is not yet in the validated cohort** (Finding F011). Because paralogous BAF subunits can share episignatures, it is biologically plausible — but unproven — that ACTL6A variants map to the BAFopathy signature.

Clinical/imaging tests: Brain MRI (structural brain anomalies, corpus-callosum abnormalities seen across CSS), developmental/neuropsychological assessment, dysmorphology exam, hand/foot radiographs for acral anomalies. No specific biomarker or laboratory abnormality is diagnostic.

Differential diagnosis: Other BAFopathies/Coffin-Siris subtypes (ARID1B, SMARCB1, SMARCA4, SMARCE1, ARID1A), Nicolaides-Baraitser syndrome (SMARCA2), ACTL6B-related disorder (BAF53b; distinct — intellectual disability with epilepsy), and other syndromic intellectual-disability disorders. Distinguishing feature: the specific ACTL6A variant on sequencing.

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11. Outcome / Prognosis

- **Survival/mortality:** No evidence of reduced life expectancy from the core neurodevelopmental phenotype; most BAFopathy patients survive into adulthood. Severe multi-organ involvement (as can occur in some CSS subtypes, e.g., SMARCB1/ARID1A) could affect prognosis, but ACTL6A-specific mortality data are unavailable.
- **Morbidity/function:** Dominated by **intellectual disability and language impairment**, with lifelong functional/educational impact and variable support needs. Acral anomalies are generally non-disabling.
- **Disease course/complications:** Potential BAFopathy-associated complications (feeding difficulties, recurrent infections, hypotonia, and — rarely — structural anomalies such as CDH or endocrine involvement) warrant surveillance, though their frequency in ACTL6A specifically is undefined.
- **Prognostic factors:** Likely variant type/location and degree of intellectual disability; formal prognostic models do not exist for this ultra-rare disorder.

- **QoL measures:** No disorder-specific instruments reported.

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12. Treatment

There is no disease-specific or disease-modifying therapy. Management is **supportive and multidisciplinary**, mirroring Coffin-Siris/BAFopathy care:

- **Developmental/rehabilitative (NCIT suggestions):** Early intervention; **speech-language therapy**, **physical therapy** (NCIT:C15402), **occupational therapy** (NCIT:C15214); special education.
- **Supportive care:** Management of feeding difficulties (nutrition support), treatment of recurrent infections, orthopedic/surveillance management of anomalies, ophthalmologic/audiologic evaluation.
- **Pharmacotherapy:** Symptom-directed only (e.g., anti-seizure medication if seizures occur; behavioral/ADHD management as needed). No pharmacogenomic guidance specific to ACTL6A.
- **Advanced therapeutics (gene/cell/RNA/targeted/immunotherapy):** None approved or in trials for this disorder.
- **Surgical/interventional:** Only for specific structural complications (e.g., CDH repair, if present).
- **Experimental/clinical trials:** No disorder-specific trials (no NCT identifiers) identified.
- **Treatment strategy:** Individualized supportive care guided by the specific phenotype; no algorithm or combination/precision regimen exists.

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13. Prevention

- **Primary prevention:** Not applicable to a de novo dominant disorder; cannot be prevented through lifestyle or environmental measures.
- **Genetic counseling (key preventive/advisory service):** For families, recurrence risk is low (de novo), but gonadal/germline mosaicism cannot be fully excluded, so empiric recurrence risk is quoted as low-but-nonzero. **Prenatal/preimplantation genetic testing** is possible for a known familial variant.

- **Secondary prevention:** Early developmental screening and intervention to optimize functional outcomes; surveillance for BAFopathy-associated complications.
- **Immunization/public-health/environmental measures:** Not applicable (routine childhood immunizations recommended as for any child; no disorder-specific vaccine).
- **Carrier/newborn screening:** Not applicable (dominant, de novo; not on newborn screening panels).

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14. Other Species / Natural Disease

- **Taxonomy / orthologs (NCBI Gene suggestions):** Human *ACTL6A* (Gene 86); mouse *Actl6a* (Gene 56456); Drosophila ortholog **Bap55**. Paralog: *ACTL6B* (BAF53b).
- **Evolutionary conservation:** Deeply conserved actin-related chromatin-remodeling subunit. Cross-species functional conservation is demonstrated: the Drosophila *Bap55* mutant dendrite-targeting phenotype is rescued by human BAF53a or BAF53b (PMID: 21284845), and the yeast/mammalian SWI/SNF core is conserved (PMID: 10078207).
- **Natural disease in other species / veterinary relevance / zoonosis:** None reported; no naturally occurring animal disease, no breed associations (no VBO entries), and no transmissibility (not applicable to a Mendelian disorder).

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15. Model Organisms

Model	Type	Key finding	PMID
Mouse ES cells (tet-inducible *Baf53a* cKO)	Cellular/ mammalian	*Baf53a* loss → p53/p21/cleaved-Caspase-3 ↑, ESC death; rescued by Baf53a **or** Baf53b (not M3 mutant)	29070872
Mouse ES cells	Cellular/ mammalian	Actl6a protects ESCs from primitive-endoderm differentiation via Nanog/Sox2	25802002
Mouse (npBAF-subunit dosage reduction)	Mammalian, in vivo	Reduced dosage → neural-tube-closure defects (spina bifida-like); BAF53a→BAF53b switch gates neurogenesis	19561591
Mouse developing cortex (BAF53a loss)	Mammalian, in vivo	BAF53a loss stalls progenitor cell cycle; Polycomb ↑, ↓accessibility at SOX2/ASCL1	33602870
Drosophila (*Bap55* mutant)	Invertebrate	Dendrite/olfactory-projection mistargeting via TIP60 complex; rescued by human BAF53a/BAF53b	21284845

Genetic model types available: Conditional/inducible knockouts and rescue constructs in mouse ESCs; Drosophila loss-of-function mutants with human-gene rescue. **No knock-in mouse carrying a patient ACTL6A missense variant (e.g., p.Arg377Trp) has been reported** — a key gap.

Phenotype recapitulation: Models robustly recapitulate the **core mechanistic axis** (progenitor proliferation, cell-cycle control, neurogenesis switch, dosage sensitivity), strongly supporting the human mechanism — e.g., Zhu et al. 2017 (PMID: 29070872): *"Baf53a is involved in the survival of ES cells by regulating p53 and Caspase3, and that Baf53b is able to compensate for this functional aspect of Baf53a."* **Limitations:** Existing models test null/knockdown rather than the specific heterozygous missense alleles seen in patients; they do not reproduce the full human syndromic phenotype (dysmorphism, acral anomalies, cognition). Resources: MGI (*Actl6a*), FlyBase (*Bap55*), Alliance of Genome Resources.

Mechanistic Model / Interpretation (Synthesis)

ACTL6A-related BAFopathy is best understood as a **dosage-sensitive chromatin-remodeling disorder of neurodevelopment**. BAF53a is the actin-related subunit that defines the **neural-progenitor** BAF complex; its tightly timed, microRNA-driven replacement by BAF53b converts proliferating progenitors into post-mitotic neurons. Germline perturbation of BAF53a dosage or function derails this switch at the point of cell-cycle exit — Polycomb accumulates, pioneer-factor (SOX2/ASCL1) sites lose accessibility, and neurogenesis is blocked — producing intellectual disability with prominent language involvement and, through BAF's broader developmental roles, the craniofacial and acral features that place the disorder within the Coffin-Siris/BAFopathy spectrum.

Three independent evidence streams converge on this model: (1) **human genetics** (recurrent de novo variants; BAFopathy-recognizable phenotype), (2) **population constraint** (pLI 0.997 / LOEUF 0.25 — the gene cannot tolerate loss), and (3) **developmental biology** (mouse/fly models proving BAF53a's progenitor-specific, cell-cycle-gating, dosage-sensitive role). The same gene's **somatic** amplification drives squamous cell carcinoma via the opposite dosage direction — a biological mirror image that reinforces the centrality of **BAF53a dosage** to cell fate. Son & Crabtree (2014, PMID: 25195934) frame the broader principle: *"The high frequency of BAF subunit mutations in neurological disorders underscores the rate-determining role of BAF complexes in neural development, homeostasis, and plasticity."*

Evidence Base

PMID	Title (abbrev.)	Role
28649782	Heterozygous ACTL6A variants → intellectual disability	**Establishes disease gene** (first description, 3 subjects)
31994175	p.Arg377Trp recognizable BAF-opathy	Recurrent variant; dysmorphism + acral malformations
34906496	BAFopathy exome cohort (16,243)	Independent de novo/pLoF support
19561591	miRNA-mediated BAF switching	Mechanism: npBAF→nBAF, dosage sensitivity
33602870	BAF subunit switching / cell-cycle exit	Mechanism: cell-cycle stall, Polycomb, SOX2/ASCL1
33408251	ACTL6A in DNA repair	Multi-complex membership (SWI/SNF, TIP60, INO80)
21284845	Bap55 / TIP60 dendrite targeting	Cross-species conservation; rescue by BAF53a/b
25169878	CSS/BAFopathy review	Defines BAFopathy/CSS core phenotype
29070872	Baf53a ESC survival	Model: p53-apoptosis, Baf53b compensation
25802002	Actl6a protects ESCs	Model: pluripotency via Nanog/Sox2
30459321	BAFopathy episignatures	Diagnostic tool (excludes ACTL6A)
38751117	DNA-methylation in NDD	Paralogous subunits share episignatures
25195934	BAF in neural development	Rate-determining role of BAF in neurodevelopment
34687603	ACTL6A in SCC	Contrasting somatic oncogenic role
34485408	BAF-opathies/SSRIDD case report	Additional ACTL6A patient; SSRIDD terminology

Limitations and Knowledge Gaps

1. **Very small patient number** (<~10 reported) → no reliable phenotype frequencies, natural history, prognosis, penetrance, or genotype-phenotype correlation.
2. **Variant classification is immature:** the recurrent literature variants (p.Arg377Trp, p.His105Arg, p.Glu227Gln) are **Conflicting** in ClinVar; most single-gene SNVs are VUS. Many ClinVar "pathogenic" entries are large 3q CNVs spanning ACTL6A, not point mutations (Finding F009).
3. **No ACTL6A-specific episignature** yet validated (Finding F011), limiting a powerful variant-reclassification route available for other BAFopathies.
4. **No OMIM/Orphanet entry**, no ICD/MeSH code, and no curated phenotype associations in Monarch (Finding F010).
5. **Mode of action (haploinsufficiency vs dominant-negative) unresolved** for patient missense variants; existing models use null/knockdown, not knock-in patient alleles.
6. **No disease-specific therapy or clinical trials.**

Proposed Follow-up Experiments / Actions

1. **Build a patient registry / GeneMatcher cohort** to aggregate additional ACTL6A cases and define phenotype frequencies, natural history, and genotype-phenotype correlations.
2. **Derive an ACTL6A DNA-methylation episignature** by including confirmed ACTL6A patients in EpiSign cohorts (leveraging paralog-sharing from PMID: 38751117) to reclassify the many VUS.
3. **Functional assays for recurrent missense variants** (p.Arg377Trp, p.His105Arg, p.Glu227Gln): BAF53a incorporation into BAF/TIP60/INO80 complexes, β-actin/BRG1 binding, and chromatin accessibility at SOX2/ASCL1 sites — to resolve LoF vs dominant-negative.
4. **Generate knock-in mouse / patient-iPSC-derived cortical organoid models** carrying p.Arg377Trp to test recapitulation of cell-cycle stalling and neurogenesis defects.
5. **Submit structured variant curation to ClinVar/ClinGen** and pursue MONDO→OMIM/Orphanet mapping to formalize the disease entity.
6. **Define surveillance guidelines** for BAFopathy-associated complications (feeding, infection, structural anomalies) adapted to ACTL6A.

Evidence source types: human clinical genetics (Marom 2017, Pascolini 2020, Chen 2022, Yuan 2021); model organism (Yoo 2009, Braun 2021, Zhu 2017, Lu 2015, Tea & Luo 2011); in vitro/biochemical (Xiao 2021, Phelan 1999); computational/database (gnomAD constraint, ClinVar, Monarch/MONDO, UniProt).